

3. PHARMACEUTICAL FORMULATION SCIENCE — DEEP DIVE (Part 2)

PHARMACEUTICAL FORMULATION SCIENCE — DEEP DIVE (Part 2)

Formulation Development Process + Solubility Enhancement + Practical Lab

Companion to Part 1 (Dosage Forms + Excipients)

For: Final Year M.Pharm (Pharmaceutics) → R&D Scientist

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PART 3 — FORMULATION DEVELOPMENT PROCESS

3.1 PRE-FORMULATION STUDIES

Important

Preformulation is the **first step** in rational formulation development. Every Indian generics company (Sun Pharma, DRL, Cipla) and MNC (Pfizer, Novartis) expects you to know this cold. Skipping preformulation = building a house without surveying the land.

Complete Preformulation Workflow

STEP 1: API Receipt & Documentation

→ Certificate of Analysis (CoA), DMF reference, structural formula

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STEP 2: Literature Review

→ Known polymorphs, published solubility data, BCS class, pKa, Log P

→ Patent landscape (salt forms, crystal forms claimed)

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STEP 3: Physical Characterization

→ Appearance, color, odor, particle size, morphology (SEM), flow properties

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STEP 4: Solid-State Characterization

→ DSC, TGA, XRPD, hot-stage microscopy → polymorph identification

→ Hygroscopicity (DVS – Dynamic Vapor Sorption)

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STEP 5: Solubility & Dissolution

→ Thermodynamic solubility (shake-flask, 24–72h equilibrium)

→ pH-solubility profile (pH 1.2, 4.5, 6.8, 7.4)

→ Kinetic solubility (DMSO dilution method)

→ Intrinsic dissolution rate (IDR, mg/cm²/min)

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STEP 6: pKa Determination

→ Potentiometric titration or UV spectrophotometric method

→ Determines ionization state → influences solubility, stability, absorption

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STEP 7: Partition Coefficient (Log P / Log D)

- Shake-flask method (octanol/water)
- Log P >3 suggests lipophilic → consider lipid formulation
- Log D at physiological pH more relevant than Log P

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STEP 8: Salt Screening (if needed)

- Counter-ions: HCl, mesylate, besylate, fumarate, maleate, tartrate, citrate, phosphate
- Criteria: Solubility improvement, stability, hygroscopicity, crystallinity, scalability
- Example: Amlodipine BESYLATE chosen over HCl for better crystallinity and stability

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STEP 9: Excipient Compatibility Studies

- Binary mixtures (API:excipient 1:1 and 1:5 w/w)
- Store at 40°C/75% RH for 2-4 weeks (accelerated)
- Analyze by: DSC (thermal events), FTIR (new peaks/shifts), XRPD (crystallinity changes)
- Visual inspection (color change, clumping)
- HPLC assay and related substances

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STEP 10: Forced Degradation / Stress Testing

- Acid (0.1N HCl, 60°C, 24h), Base (0.1N NaOH, 60°C, 24h)
- Oxidative (3% H₂O₂, RT, 24h), Thermal (60°C solid, 7 days)

- Photolytic (ICH Q1B conditions: 1.2M lux·h + 200 W·h/m² UV)
- Humidity (75% RH, 40°C, 7 days)
- Goal: 5–20% degradation to identify degradation pathways
- Develop stability-indicating HPLC method that resolves all degradants

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STEP 11: BCS Classification

- Solubility: Highest dose dissolves in ≤250 mL at pH 1–6.8 → HIGH
- Permeability: ≥85% absorbed (or Papp > metoprolol in Caco-2) → HIGH
- BCS Class determines formulation strategy:

BCS-Driven Formulation Strategy

BCS Class	Solubility	Permeability	Formulation Strategy	Example Drug
I	High	High	Simplest — standard IR tablet; biowaiver possible	Metoprolol, Propranolol
II	Low	High	Solubility enhancement critical — ASD, SEDDS, nanosizing, salt form	Itraconazole, Fenofibrate, Ibuprofen
III	High	Low	Permeation enhancement; rapid dissolution needed; biowaiver possible	Metformin, Acyclovir
IV	Low	Low	Most challenging — combination of solubility + permeability enhancement	Furosemide, Taxol

Complete Preformulation Checklist

#	Parameter	Method	Target / Output
1	Molecular weight & structure	Literature / CoA	Document; identify ionizable groups

#	Parameter	Method	Target / Output
2	Appearance & organoleptic	Visual, taste	Color, form, odor, taste
3	Melting point / decomposition	DSC, capillary method	°C; sharp = crystalline; broad = amorphous/impure
4	Polymorphic forms	DSC, XRPD, hot-stage microscopy	Identify all forms; determine thermodynamic stability
5	Crystallinity vs amorphous	XRPD (sharp peaks = crystalline)	% Crystallinity
6	Hygroscopicity	DVS or humidity chamber	Classify: non-hygroscopic (<0.2% at 80% RH) to very hygroscopic
7	Particle size distribution	Laser diffraction (Malvern), sieve analysis	D10, D50, D90, span
8	Particle morphology	SEM	Shape (needle, plate, prismatic) → impacts flow and compressibility
9	Bulk / Tapped density	USP <616>	Calculate Carr's Index, Hausner Ratio
10	Flow properties	Angle of repose, Carr's Index	Assess processability
11	True density	Helium pycnometer	g/cm ³
12	pKa	Potentiometric or spectrophotometric	Ionization behavior
13	Solubility (aqueous)	Shake-flask (24–72h, 37°C)	mg/mL at pH 1.2, 4.5, 6.8, 7.4; classify HIGH/LOW per BCS
14	Solubility in co-solvents	Shake-flask	PEG 400, propylene glycol, ethanol, DMSO
15	Log P / Log D	Shake-flask (octanol/water)	Lipophilicity indicator
16	Intrinsic Dissolution Rate	Wood's apparatus (USP)	mg/cm ² /min — predict in vivo dissolution
17	Salt screening	Counter-ion library	Rank by solubility, crystallinity, hygroscopicity, stability
18	Excipient compatibility	Binary mixtures, DSC/FTIR/HPLC at 40°C/75% RH	Compatible / Incompatible for each excipient
19	Forced degradation	Acid, base, oxidation, heat, light, humidity	Degradation pathways; stability-indicating method

#	Parameter	Method	Target / Output
20	Solution stability (pH)	Buffer solutions at pH 1–9, 37°C, 24h	Identify pH of maximum stability
21	Photostability	ICH Q1B light exposure	Light-sensitive? → Amber container, opaque coating
22	BCS Classification	Dose-solubility ratio + permeability data	Class I, II, III, or IV

3.2 FORMULATION SCREENING AND OPTIMIZATION (QbD/DoE)

Step-by-Step QbD Framework

STEP 1: Define QTPP (Quality Target Product Profile)

→ Dosage form, route, strength, release profile, stability, packaging

Example: "SR Tablet, Oral, 500mg Metformin HCl, 8–12h release, 24-month shelf life"

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STEP 2: Identify CQAs (Critical Quality Attributes)

→ Drug product attributes that affect safety/efficacy

→ For SR tablet: Dissolution profile (CQA#1), Assay (CQA#2), Content Uniformity (CQA#3), Related Substances (CQA#4)

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STEP 3: Risk Assessment – Identify CPPs and CMAs

→ Use Ishikawa (fishbone) diagram: Materials → Methods → Machines → Measurement → Man → Environment

→ Use FMEA (Failure Mode and Effects Analysis): Rate Severity × Occurrence × Detection → RPN score

→ High RPN factors = candidates for DoE study

→ Example CPPs: HPMC concentration, compression force, granule moisture

→ Example CMAs: API particle size, HPMC viscosity grade, MCC grade

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STEP 4: Screening DoE (if >5 factors)

→ Plackett-Burman design: Screen 7-11 factors in 12 runs

→ Identifies which factors are STATISTICALLY SIGNIFICANT

→ Output: Ranked list of critical factors → select top 3-4 for optimization

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STEP 5: Optimization DoE

→ With 2-4 critical factors:

- 2 factors → Full factorial (3^2) = 9 runs + center points

- 3 factors → Box-Behnken (15 runs) or CCD (20 runs)

- 4+ factors → D-optimal or definitive screening design

→ Responses: Dissolution at 1h, 4h, 8h; Hardness; Friability

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STEP 6: Model Building & Analysis

→ Fit polynomial model (linear, quadratic, interaction terms)

→ Check: R^2 , Adjusted R^2 , Predicted R^2 , Lack of Fit p-value (>0.05 = good)

→ Generate: Pareto charts, contour plots, 3D surface plots

→ Identify optimal region (desirability function for multiple responses)

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STEP 7: Establish Design Space

→ Region of CPP/CMA ranges where ALL CQAs are simultaneously met

- Edges of design space = proven acceptable ranges (PARs)
- Working within design space ≠ regulatory change (ICH Q8 advantage)

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STEP 8: Verification Runs

- Run 3–6 batches at predicted optimum + edge of design space
- Confirm that model predictions match actual results
- If yes → Design Space is validated → proceed to scale-up

Worked Example: QbD Optimization of Metformin 500mg SR Tablet

QTPP:

- Oral tablet, 500mg, sustained release over 8 hours
- Dissolution: 20–40% at 1h, 40–65% at 4h, NLT 80% at 8h
- Stability: 24 months at 30°C/65% RH (Zone IVa)

CQAs: Dissolution at 1h (Y1), 4h (Y2), 8h (Y3); Hardness (Y4)

Risk Assessment → Critical Factors:

1. **X1: HPMC K100M concentration (%)** — 25%, 30%, 35%
2. **X2: Compression force (kN)** — 10, 15, 20
3. **X3: MCC PH-102 concentration (%)** — 15%, 20%, 25%

DoE Selected: Box-Behnken Design (3 factors, 3 levels) → **15 runs + 3 center points = 18 total**

Key Results (Example):

- HPMC concentration is the **most significant factor** ($p < 0.001$) for all dissolution responses
- Compression force has a **significant interaction** with HPMC% on dissolution at 1h
- MCC% has minimal effect on dissolution but significantly affects hardness
- Optimum: HPMC 30%, Compression 15 kN, MCC 22% → gives: 30% at 1h, 55% at 4h, 88% at 8h, Hardness 12 kp

Design Space: HPMC 28–33%, Compression 12–18 kN, MCC 18–25%

Verification: 3 batches at center + 3 at edges → all within specification → Design Space validated

Interview Q: "Walk me through how you applied QbD in your thesis."

Framework: (1) State your QTPP. (2) List your CQAs. (3) Describe your risk assessment approach. (4) Name the DoE design and WHY you chose it. (5) State the key finding — which factor was most significant and what was the optimal setting? (6) Show awareness of design space concept. Even if your thesis didn't formally use QbD, frame it retrospectively: "If I were to apply QbD, I would define QTPP as... and use Box-Behnken because I had 3 key factors."

3.3 SCALE-UP CONSIDERATIONS

Scale Definitions

Scale	Batch Size	Equipment	Purpose
Lab	100–500g	Mortar/pestle, small V-blender, single-punch press	Feasibility, screening
Pilot	5–50 kg	Pilot RMG (10L), pilot FBD, rotary press (D-tooling)	Optimization, stability batches, exhibit batches
Commercial	100–500+ kg	Production RMG (300–600L), production FBD, high-speed rotary press (B-tooling)	Registration, commercial supply

Scale-Up Rules

Granulation (RMG — Rapid Mixer Granulator):

Parameter	Scale-Up Principle	Formula
Impeller Tip Speed	Keep CONSTANT across scales	Tip Speed = $\pi \times D \times N$ (m/s)
Froude Number	Dimensionless; keep constant	$Fr = N^2 \times R / g$
Specific Power Consumption	Monitor power curve → endpoint determination	kW/kg of blend

Parameter	Scale-Up Principle	Formula
Granulation Time	NOT directly proportional → adjust empirically	—
Binder Addition Rate	Scale proportionally to batch size	mL/min per kg

Example: Lab RMG (2L bowl, 10cm impeller): If impeller speed = 300 RPM → Tip Speed = $\pi \times 0.10 \times 300/60 = 1.57 \text{ m/s}$. For pilot RMG (10L, 25cm impeller): Target same tip speed 1.57 m/s → $N = 1.57 \times 60 / (\pi \times 0.25) = \textbf{120 RPM}$

Compression (Tablet Press):

Parameter	Lab Press	Production Press	Key Consideration
Press Type	Single-punch or small rotary	High-speed rotary (Fette 2090, Korsch XL)	Different compression profiles
Dwell Time	Long (20–50 ms)	Short (5–15 ms at high speed)	Shorter dwell → less bonding → may need reformulation
Compression Force	Easy to control	Must replicate lab force profile	Use force-displacement curve matching
Tooling	D-tooling (small)	B-tooling (standard production)	Different punch face area → recalculate force
Speed	10–20 RPM	40–80 RPM	Speed sensitivity must be evaluated

Coating (Pan Coater):

Parameter	Scale-Up Approach
Pan loading	60–70% of pan volume at both scales
Spray rate	Scale by: $\text{Spray Rate} \propto (\text{Pan Diameter})^2 \times \text{Pan Speed}$
Inlet air volume	Scale proportionally to batch size; maintain same air velocity
Pan speed	Maintain same linear velocity at tablet bed surface
Gun-to-bed distance	Keep constant (15–25 cm) or scale by pan diameter ratio

Common Scale-Up Failures

Failure	Root Cause	Solution
Over-granulation at scale	Longer binder addition time in larger granulator	Optimize spray rate; use impeller torque/power endpoint
Capping at production speed	Shorter dwell time → insufficient bonding	Pre-compression force; slower speed; reformulate with more plastic excipient (MCC)
Weight variation	Different flow pattern in production press hopper	Use force-feeder; optimize granule PSD
Dissolution failure	Different mixing intensity for lubricant	Control Mg stearate blending time precisely; use shear-sensitive formulation approach
Coating defects (orange peel, logo bridging)	Different spray dynamics at scale	Adjust spray rate, atomization air, gun-to-bed distance

3.4 TECHNOLOGY TRANSFER

What Gets Transferred

R&D Site (Sending Unit) —————> Manufacturing Site (Receiving Unit)

Documents transferred:

- └─ Master Batch Record (MBR) / Master Formula Record (MFR)
- └─ Specifications (API, excipients, in-process, finished product)
- └─ Analytical Methods (validated, with method transfer protocol)
- └─ Stability Data (from exhibit/pivotal batches)
- └─ Process Validation Protocol
- └─ Cleaning Validation Protocol
- └─ Equipment Qualification Requirements
- └─ Critical Process Parameters (CPPs) with proven acceptable ranges

└─ Control Strategy Document

└─ Regulatory Filing Dossier (Module 3.2.P sections)

Key Documents

Document	Purpose
Technology Transfer Protocol	Defines scope, responsibilities, acceptance criteria
Master Batch Record	Step-by-step manufacturing instructions with CPPs
Analytical Method Transfer Protocol	Procedure to transfer validated methods to receiving lab
Process Validation Protocol	3 consecutive successful batches at commercial scale
Stability Protocol	Conditions, timepoints, test parameters for registration stability

3.5 DoE APPLICATION — COMPLETE WORKED EXAMPLE

Scenario: Optimize an IR Tablet for Poorly Soluble Drug (BCS Class II)

Objective: Maximize dissolution (Q30 $\geq 80\%$) while maintaining acceptable hardness (4–8 kp) and friability (NMT 1%)

Factors Identified (from risk assessment):

Factor	Type	Low (-1)	High (+1)
X1: SLS concentration (%)	Numeric	0.5	2.0
X2: Croscarmellose sodium (%)	Numeric	2.0	6.0
X3: Compression force (kN)	Numeric	8	16

Design: Full factorial 2^3 with center points = **8 corners + 3 center points = 11 runs**

Responses: Y1 = Dissolution at 30 min (%); Y2 = Hardness (kp); Y3 = Disintegration time (sec)

Analysis in JMP/Design-Expert:

1. Run all 11 batches → collect response data
2. Fit response surface model: $Y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_{12}X_1X_2 + \beta_{13}X_1X_3 + \beta_{23}X_2X_3 + \epsilon$
3. **ANOVA table**: Check p-values for each term
 - If $p < 0.05$ → factor is significant
 - If $p > 0.10$ → factor can be removed from model
4. **Pareto chart**: Bars show standardized effect size → identify which factors matter most
5. **Contour plot**: 2D map showing response as function of 2 factors → find sweet spot
6. **Desirability function**: Optimize all 3 responses simultaneously → JMP gives optimal point

Typical Result:

- SLS% is the strongest predictor of dissolution ($p < 0.001$)
- CCS% significantly affects disintegration time ($p = 0.003$)
- Compression force primarily affects hardness ($p < 0.001$) with minor effect on dissolution
- Optimal: SLS 1.5%, CCS 4%, Compression 12 kN → Q30 = 89%, Hardness = 6.2 kp, DT = 180 sec

How to Present DoE in Interview

"In my thesis, I used a **Box-Behnken design with 3 factors at 3 levels** to optimize my sustained-release tablet. My independent variables were HPMC K100M concentration (20-40%), compression force (10-20 kN), and MCC amount (10-30%). My responses were dissolution at 1h, 4h, and 8h, plus hardness. I used **Design-Expert 13** to analyze the data. The ANOVA showed that HPMC concentration was the most significant factor ($p < 0.0001$) affecting all dissolution timepoints. There was a significant interaction between HPMC% and compression force on the 1h dissolution ($p = 0.015$). My optimized formulation achieved $f_2 > 55$ with the innovator product. The design space was HPMC 28-35%, compression 12-18 kN."

3.6 STABILITY STUDY DESIGN (ICH Q1A)

Storage Conditions by Climatic Zone

Zone	Climate	Long-Term	Accelerated	Intermediate
I	Temperate	21°C/45% RH	40°C/75% RH	—
II	Mediterranean/Subtropical	25°C/60% RH	40°C/75% RH	30°C/65% RH

Zone	Climate	Long-Term	Accelerated	Intermediate
III	Hot/Dry	30°C/35% RH	40°C/75% RH	—
IVa	Hot/Humid (India)	30°C/65% RH	40°C/75% RH	—
IVb	Hot/Very Humid (Brazil, ASEAN)	30°C/75% RH	40°C/75% RH	—

Important

India falls in Zone IVa. Long-term storage: **30°C/65% RH**. This is stricter than USA/EU (Zone II: 25°C/60% RH). If a product passes Zone IVa stability, it automatically passes Zone II.

Testing Intervals

Study Type	Duration	Timepoints
Long-term	12 months minimum (24–36 months for full shelf-life)	0, 3, 6, 9, 12, 18, 24, 36 months
Accelerated	6 months	0, 1, 2, 3, 6 months
Intermediate	12 months (if accelerated fails)	0, 3, 6, 9, 12 months

Shelf-Life Determination Rules

Accelerated Result	Action
No significant change at 6 months	Shelf life = 2× long-term data. With 12-month LT data → propose 24-month shelf life
Significant change at 3–6 months	Conduct intermediate study (30°C/65% RH); shelf life based on LT data only
Significant change before 3 months	Product is unstable under accelerated conditions; may need reformulation

"Significant Change" defined as (ICH Q1A):

- 5% change in assay from initial
- Any degradation product exceeding acceptance criterion
- Failure to meet dissolution specification
- Failure of appearance, physical attributes, or functionality test
- Failure of pH outside specified range

- Failure to meet water content specification

Stability Protocol Template

STABILITY PROTOCOL

Product: [Name], [Strength], [Dosage Form]

Batch Number: [XXX] Batch Size: [XX kg] Mfg Date: [DD/MM/YYYY]

Pack: [HDPE bottle with CRC / Alu-Alu blister / PVC-PVDC blister]

Storage Conditions:

Long-term: 30°C ± 2°C / 65% RH ± 5% RH (India Zone IVa)

Accelerated: 40°C ± 2°C / 75% RH ± 5% RH

Testing Intervals:

Long-term: 0, 3, 6, 9, 12, 18, 24, 36 months

Accelerated: 0, 1, 2, 3, 6 months

Parameters to Test:

Parameter	Method	Spec	Timepoints
Description	Visual	[X]	All
Assay	HPLC	[X]	All
Related Substances	HPLC	[X]	All

Dissolution	USP App II	[X]	All	
Water Content	Karl Fischer	[X]	All	
Hardness	Hardness tstr	[X]	0, 6, 12, 24, 36	
Microbial Limits	USP <61>/<62>	[X]	0, 12, 24, 36	

Common Stability Failures and Root Cause

Failure	Possible Root Cause	Investigation
Assay drop >5%	Degradation (hydrolysis, oxidation); drug-excipient interaction	Run forced degradation; check HPLC for new peaks; compare with compatibility data
Increase in impurities	Oxidative degradation; residual peroxides in excipients (PVP, PEG)	Add antioxidant (BHT); switch PVP grade; check peroxide levels in excipients
Dissolution slow-down	Cross-linking of gelatin capsule shell; hardening of tablet on aging	Switch to HPMC capsule; add SLS to dissolution medium; investigate excipient interactions
Discoloration	Maillard reaction (lactose + amine); oxidation; photodegradation	Remove lactose; add antioxidant; use opaque packaging; amber blister
Moisture increase	Hygroscopic API or excipient; inadequate packaging	Use desiccant; switch to Alu-Alu blister; use less hygroscopic excipient

PART 4 — SOLUBILITY ENHANCEMENT TECHNIQUES

~40% of marketed drugs and ~90% of drugs in development pipeline are poorly water-soluble (BCS Class II/IV). Solubility enhancement is one of the **hottest skills** in pharma R&D. This section covers all 15 major industrial techniques.

Technique 1: Micronization & Nanosizing

When to Use

- BCS Class II drugs where dissolution rate is the rate-limiting step
- Drug has adequate stability in crystalline form
- Solubility limited by surface area, not intrinsic solubility

Mechanism

Noyes-Whitney equation: $dC/dt = D \cdot A \cdot (C_s - C) / h$

- Reducing particle size → increases **A (surface area)** → increases dissolution rate
- Nanosizing also increases **C_s (saturation solubility)** via Ostwald-Freundlich effect (for particles <1 µm)

Methods

Method	Output Size	Equipment	Industry Use
Jet milling (micronization)	2–10 µm	Air jet mill (Jet-O-Mizer)	Very common for API micronization
Wet media milling	100–400 nm (nanocrystals)	NanoMill (Elan), Dyno-Mill	Rapamune (Sirolimus), Emend (Aprepitant), Megace ES (Megestrol)
High-pressure homogenization (HPH)	100–500 nm	Microfluidizer, GEA Panda	Triglide (Fenofibrate)
Dry milling	5–50 µm	Ball mill, hammer mill	Routine API sizing

Marketed Examples

Product	Drug	Technology	Size
Rapamune (Pfizer)	Sirolimus	NanoCrystal (wet milling)	~200 nm
Emend (Merck)	Aprepitant	NanoCrystal	~120 nm
Megace ES	Megestrol	NanoCrystal	~150 nm
Triglide	Fenofibrate	IDD-P (HPH)	~300 nm

| Invega Sustenna | Paliperidone palmitate | Wet milling (IM nanocrystal) | ~300 nm |

Interview Q: "What is the Ostwald-Freundlich equation and why does nanosizing increase saturation solubility?"

As particle size decreases below ~1 µm, the curvature of the particle surface increases, leading to higher surface energy and increased dissolution pressure. The Ostwald-Freundlich equation: $\ln(C_s/C_\infty) = 2\gamma V / rRT$, where γ = interfacial tension, V = molar volume, r = particle radius. Below 200–300 nm, the effect becomes significant — saturation solubility can increase 10–50%. However, nanocrystals are thermodynamically unstable (Ostwald ripening) and need stabilizers (polymers like HPMC, surfactants like SLS, Tween 80) to prevent crystal growth during storage.

Technique 2: Salt Formation

When to Use

- API has an ionizable group (acid or base)
- Free form has poor solubility
- **First-line strategy** — simplest, most scalable approach

Common Salt Forms

API Type	Counter-ion Examples	Marketed Example
Weak Base	HCl, mesylate, besylate, maleate, fumarate, tartrate, citrate, phosphate, sulfate	Metformin HCl , Amlodipine besylate , Imatinib mesylate (Gleevec)
Weak Acid	Sodium, potassium, calcium, meglumine, tromethamine, lysine	Diclofenac sodium , Naproxen sodium , Atorvastatin calcium

Selection Criteria (Salt Screening)

Criterion	What to Evaluate
Solubility improvement	Must show ≥ 3 – $10\times$ improvement over free form
Crystallinity	Must form a well-defined crystalline salt (amorphous salts = stability risk)
Hygroscopicity	Low hygroscopicity preferred for solid dosage
Stability	Accelerated stability (40°C/75% RH, 4 weeks)
Bioavailability	Salt must dissolve AND maintain supersaturation → may need to test in vivo
Scalability	Reaction must be scalable and reproducible
IP (Intellectual Property)	Is the salt form patented? Check freedom-to-operate
Regulatory precedence	Is the counter-ion in FDA IIG database?

Interview Q: "Why was besylate chosen for Amlodipine instead of HCl?"

Amlodipine HCl salt exists but is **hygroscopic and poorly crystalline**, making it difficult to process into a stable tablet. Amlodipine besylate (benzenesulfonate salt) gives a well-defined crystalline form with excellent stability, low hygroscopicity, and good aqueous solubility. The Pfizer patent specifically covers the besylate salt form — this was both a scientific AND strategic (IP) decision.

Technique 3: Amorphous Solid Dispersions (ASD)

When to Use

- BCS Class II drugs with high melting point and low solubility
- Salt formation insufficient or not possible (non-ionizable drug)
- **Industry trend: Fastest-growing solubility platform in pharma R&D**

Mechanism

- Drug is dispersed in an amorphous state within a polymer matrix

- Amorphous form has **higher free energy** → higher apparent solubility (10–1600× improvement)
- Polymer **inhibits crystallization** during storage and in GI tract (maintains supersaturation = "parachute" effect)

Key Polymers for ASD

Polymer	Tg (°C)	Solubility	Best Manufacturing Method	Key Advantage
PVP-VA (Copovidone, Kollidon VA 64)	101	Water-soluble	HME (most common)	Good processability; lower Tg than PVP
HPMC-AS (Affinisol, Aqoat)	120	pH-dependent (dissolves >pH 5.5)	Spray drying (preferred)	Excellent supersaturation maintenance; "spring & parachute"
Soluplus (PVA-PEG graft copolymer)	70	Water-soluble	HME	Low processing temperature; acts as solubilizer
PVP K30 (Povidone)	149	Water-soluble	Spray drying, co-precipitation	High Tg for stability; but very hygroscopic
HPMC E50 (Methocel)	160–210	Water-soluble	Spray drying	Available but less used than HPMC-AS for ASD
Eudragit L100	>150 (decomposes)	pH-dependent	Spray drying	Enteric + ASD combination

Manufacturing Methods

Method	Process	Advantages	Limitations	Marketed Example
Hot-Melt Extrusion (HME)	API + polymer melted/mixed in twin-screw extruder → extrudate → milled → compressed	Continuous; no solvents; scalable; good mixing	High temperature may degrade heat-labile APIs; limited to	Kaletra (Lopinavir/Ritonavir) — Abbvie; Noxafil (Posaconazole) — Merck; Viekira Pak — Abbvie

Method	Process	Advantages	Limitations	Marketed Example
			thermoplastic polymers	
Spray Drying	Drug + polymer dissolved in organic solvent → atomized into hot gas → rapid solvent evaporation → amorphous powder	Low process temperature; works with HPMC-AS; fine particle size	Requires organic solvent handling; lower throughput; scale-up of spray dryer is complex	Intelence (Etravirine) — J&J; Zelboraf (Vemurafenib) — Roche; Kalydeco (Ivacaftor) — Vertex
Co-precipitation	Drug + polymer dissolved → anti-solvent addition → precipitation	Lab-scale simple	Scale-up difficult; inconsistent	Research stage mostly

Interview Q: "Explain the 'spring and parachute' concept."

Model Answer: In amorphous solid dispersions, the "spring" is the rapid supersaturation generated when the amorphous drug dissolves faster than crystalline form — the drug concentration temporarily exceeds thermodynamic solubility. The "parachute" is the polymer's role in **maintaining this supersaturation** by inhibiting drug crystallization from the supersaturated solution. HPMC-AS is the best "parachute" polymer because it adsorbs onto drug crystal nuclei, preventing crystal growth. Without the parachute, the drug would rapidly crystallize (precipitation), and you lose the solubility advantage. The goal: spring high + parachute long enough for absorption to occur.

Technique 4: Cyclodextrin Complexation

Mechanism

- Cyclodextrins are **cyclic oligosaccharides** with a hydrophobic cavity and hydrophilic exterior
- Drug molecule (guest) enters the hydrophobic cavity → forms an **inclusion complex** → dramatically improves aqueous solubility
- Complex is in equilibrium: $\text{Drug} + \text{CD} \rightleftharpoons \text{Drug} \cdot \text{CD complex}$

Types Used in Pharma

Cyclodextrin	Abbreviation	Key Property	Marketed Example
Hydroxypropyl- β -CD	HP- β -CD (Kleptose HPB, Cavitron)	Oral and parenteral safe; most widely used	Sporanox Oral Solution (Itraconazole); Vfend Oral (Voriconazole)
Sulfobutylether- β -CD	SBE- β -CD (Captisol)	Parenteral-safe; anionic	Vfend IV (Voriconazole); Kyprolis (Carfilzomib); Remdesivir (Veklury)
β -CD	—	Parent CD; limited solubility; renal toxicity risk (IV)	Topical/oral only
γ -CD (Sugammadex)	—	Larger cavity for bulkier molecules	Bridion (Sugammadex) — reversal of neuromuscular blockade

Interview Q: "What is the difference between HP- β -CD and SBE- β -CD? When would you use each?"

HP- β -CD is the standard for oral formulations — well-tolerated, good complexation for many drugs, approved in many oral products. SBE- β -CD (Captisol) was specifically developed for **IV/parenteral use** — it has a negative charge from sulfobutyl groups that prevents nephrotoxicity (unlike parent β -CD which crystallizes in renal tubules). For parenteral voriconazole (Vfend IV), Captisol was essential because HP- β -CD wouldn't provide sufficient complexation at the required concentration. Cost-wise, SBE- β -CD is more expensive, so for oral products, HP- β -CD is preferred unless SBE- β -CD gives significantly better complexation.

Technique 5: Lipid-Based Drug Delivery Systems (LBDDS)

Classification (Pouton's Lipid Formulation Classification System — LFCS)

Type	Composition	Dispersion	Example System
Type I	Oils (triglycerides) only	Crude emulsion; requires digestion	Simple lipid solution

Type	Composition	Dispersion	Example System
Type II	Oils + water-insoluble surfactant (HLB <12)	SEDDES; self-emulsifying → coarse emulsion	—
Type IIIA	Oils + water-soluble surfactant + co-solvent	SMEDDS; self-micro-emulsifying → fine emulsion (<250nm)	—
Type IIIB	Low oils + high surfactant + co-solvent	SNEDDS; self-nano-emulsifying → nanoemulsion (<100nm)	Neoral (Cyclosporine)
Type IV	Surfactant + co-solvent (no oil)	Micellar solution	—

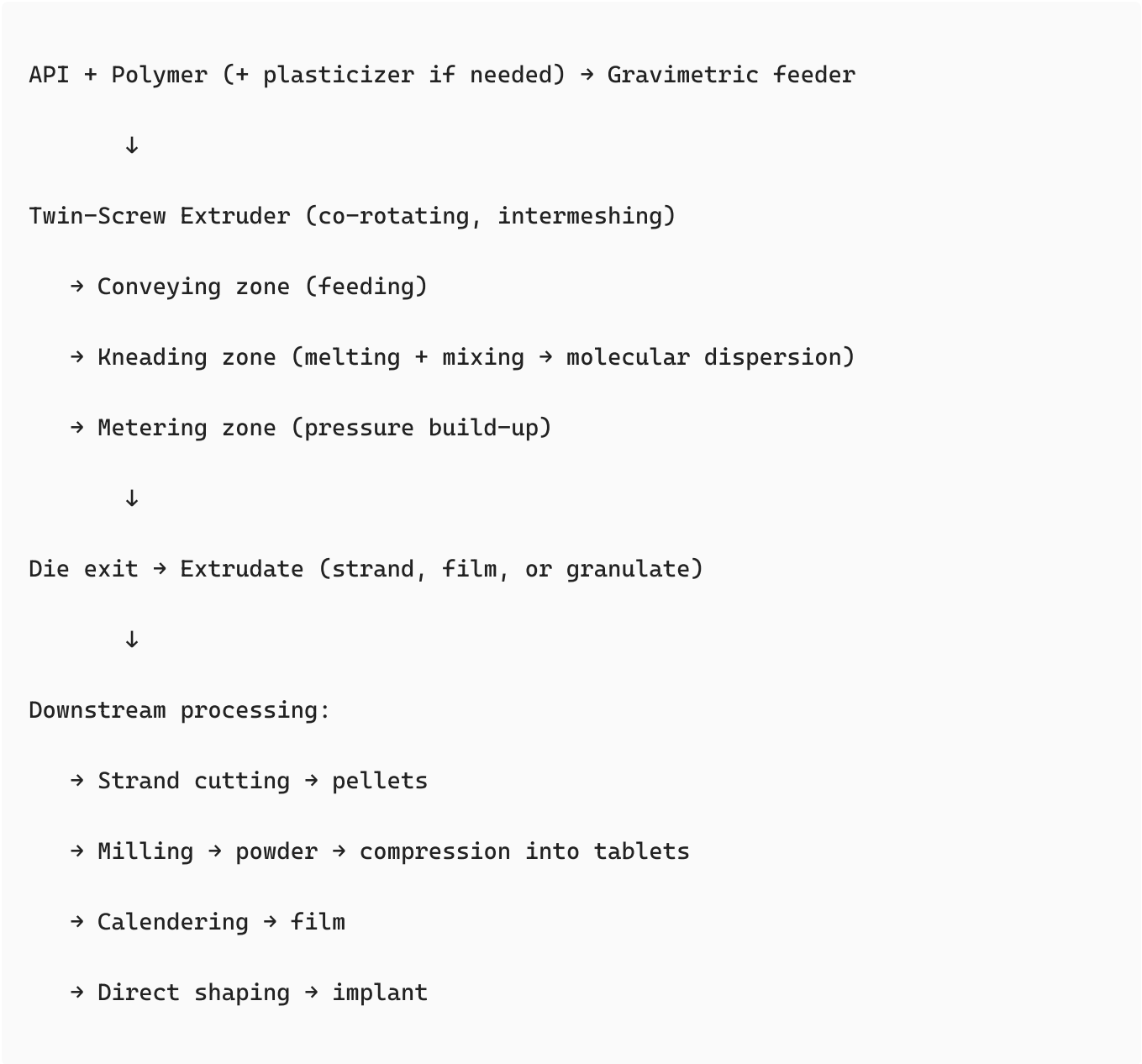
Marketed Examples

Product	Drug	LBDDS Type	Dosage Form
Neoral (Novartis)	Cyclosporine A	SNEDDS (Type IIIB)	Soft gelatin capsule
Norvir (AbbVie)	Ritonavir	Type IIIA	Soft gelatin capsule (original); now tablet (ASD)
Fortovase	Saquinavir	SEDDES (Type II)	Soft gelatin capsule
Sandimmune	Cyclosporine A	Type I (oil solution)	Soft gelatin capsule
Accutane / Absorica	Isotretinoin	Lipid solution	Soft gelatin capsule

Key Excipients for LBDDS

Category	Example	HLB
Oils (long-chain triglycerides)	Soybean oil, olive oil, corn oil	—
Oils (medium-chain triglycerides)	Miglyol 812, Captex 355	—
Surfactants	Cremophor EL (HLB 12-14), Tween 80 (HLB 15), Labrasol (HLB 14)	12–15
Co-surfactants / co-solvents	PEG 400, Transcutol HP, propylene glycol	—

Technique 6: Hot Melt Extrusion (HME) — Detailed Process



Critical Process Parameters

Parameter	Typical Range	Effect
Barrel temperature	Tg of polymer + 20–40°C	Too low → incomplete mixing; too high → degradation
Screw speed	50–200 RPM	Higher → more shear → better mixing but more heat
Feed rate	0.5–5 kg/h (lab); 10–100+ kg/h (commercial)	Affects residence time

Parameter	Typical Range	Effect
Screw configuration	Mix of conveying elements + kneading blocks	Kneading blocks provide mixing/dispersion
Residence time	1–5 minutes typically	Too long → degradation

HME Advantages Over Spray Drying

- **No organic solvents** → no residual solvent concern (ICH Q3C)
- **Continuous process** → preferred by regulatory agencies
- **Higher throughput** → better for commercial scale
- **Better for thermoplastic polymers** (PVP-VA, Soluplus, HPC)

Interview Q: "Why was HME chosen for Kaletra (lopinavir/ritonavir)?"

Original Kaletra was a soft gelatin capsule requiring refrigeration — major patient compliance issue in resource-limited settings (HIV treatment in Africa). AbbVie reformulated using **HME with PVP-VA (Kollidon VA 64)** to create amorphous solid dispersion tablets (Meltrex technology). Benefits: (1) Room-temperature stable — no refrigeration; (2) Smaller tablet — better compliance; (3) No food effect; (4) 200mg lopinavir/50mg ritonavir per tablet. This is a landmark example of HME enabling global health impact.

Technique 7: Co-Crystals

- **Definition:** Crystalline material composed of API + co-former in a defined stoichiometric ratio, held together by non-covalent interactions (H-bonding)
- **Advantage over amorphous:** Thermodynamically stable (crystalline) → better long-term stability
- **Regulatory:** FDA classifies co-crystals as **drug product intermediates** (not new APIs) since 2018
- **Example: Entresto (sacubitril/valsartan)** — a co-crystal/salt complex. **Lexapro (escitalopram oxalate)** is sometimes discussed in co-crystal context

Techniques 8–15: Summary Table

#	Technique	Mechanism	Best For	Key Marketed Example
8	pH Modification	Add acid/base to microenvironment → local solubility increase	Weakly basic drugs (need low pH for dissolution)	Buffered aspirin; Didanosine + antacid buffer
9	Surfactant Micellar Solubilization	Drug partitions into surfactant micelles above CMC	Lipophilic drugs; parenteral formulations	Taxol (Cremophor EL + ethanol); Doxil (surfactant in liposomes)
10	Phospholipid Complexation (Phytosome)	Drug complexes with phosphatidylcholine → improved permeation	Polyphenols, herbal extracts with low oral BA	Siliphos (Silybin phytosome) — Indena
11	Spray Drying	Rapid solvent evaporation → amorphous particles	Heat-labile APIs; HPMC-AS-based ASDs	Zelboraf (Vemurafenib) — Roche
12	HME	Melt mixing → molecular dispersion in polymer	Thermally stable APIs; PVP-VA/Soluplus	Kaletra, Noxafil, Viekira Pak
13	KinetiSol	High-speed rotational processing → frictional heating → ASD	Thermally labile APIs (lower temp than HME)	Research/clinical stage
14	Mesoporous Silica	Drug adsorbed into nano-pores (2–50nm) of silica → stabilized amorphous	Extremely insoluble drugs; controlled loading	Neusilin, Syloid — as carriers
15	Supersaturable Formulations	"Spring + Parachute" → generate and maintain supersaturation	Any BCS II; combine with precipitation inhibitor	All ASD products inherently use this

PART 5 — PRACTICAL LAB TASKS

5.1 Experiments to Do in University Lab RIGHT NOW

#	Experiment	Equipment	Data to Collect	CV Output
1	API Preformulation Profiling — Solubility at pH 1.2, 4.5, 6.8; melting point (DSC); FTIR spectrum	UV-Vis, DSC, FTIR, pH meter	pH-solubility profile graph; DSC thermogram; FTIR spectrum	"Conducted preformulation characterization including pH-solubility profiling and thermal analysis"
2	Excipient Compatibility Study — Binary mixtures (API + 5 excipients) at 40°C/75% RH, 4 weeks	Stability chamber, DSC, FTIR, HPLC	Compatibility table (compatible/incompatible); overlay DSC thermograms	"Performed excipient compatibility screening using DSC, FTIR, and accelerated stability"
3	IR Tablet by Direct Compression — Formulate 3 batches with varying disintegrant levels	V-blender, single-punch press, dissolution apparatus	Weight variation, hardness, friability, disintegration time, dissolution profiles	"Developed IR tablets via direct compression; optimized disintegrant concentration for >85% dissolution in 30 min"
4	IR Tablet by Wet Granulation — Compare with DC formulation	RMG (or mortar), FBD/oven, press	Granule properties (flow, LOD, PSD); tablet properties; comparison with DC	"Evaluated wet granulation vs direct compression for a poorly flowing API"

#	Experiment	Equipment	Data to Collect	CV Output
5	Sustained-Release Tablet — HPMC K100M matrix at 3 levels (20%, 30%, 40%)	Press, dissolution apparatus (12h test)	12h dissolution profiles; kinetics (zero-order, Higuchi, Korsmeyer-Peppas); f2 values	"Designed sustained-release HPMC matrix tablets; applied dissolution kinetics modeling (Higuchi, Korsmeyer-Peppas)"
6	DoE Optimization — 2-factor design on your best formulation	JMP/Design-Expert, press, dissolution	Contour plots; optimal point; design space	"Applied Box-Behnken DoE to optimize formulation parameters; established design space using JMP"
7	HPLC Method Development — Develop and validate for your API	HPLC system	Linearity ($R^2 > 0.999$), precision (RSD $< 2\%$), accuracy (98–102%), specificity	"Developed and validated HPLC method per ICH Q2(R2) for quantification of [API name]"
8	Forced Degradation Study — Acid, base, oxidation, heat, light	HPLC, stability chamber	Degradation % for each condition; chromatograms showing degradant peaks	"Conducted forced degradation studies; developed stability-indicating HPLC method"
9	Dissolution Profile Comparison — Your formulation vs innovator (f1/f2)	Dissolution apparatus, UV/HPLC	f1 (< 15) and f2 (> 50) values; overlay dissolution curves	"Demonstrated dissolution profile similarity with innovator product (f2 = 62)"
10	Film Coating Trial — Coat IR tablets with Opadry	Lab-scale pan coater or coating pan	Weight gain %, appearance, dissolution comparison (coated vs uncoated)	"Performed film coating using HPMC-based system (Opadry);

#	Experiment	Equipment	Data to Collect	CV Output
				achieved uniform 3% weight gain"
11	Stability Study Setup — Place tablets at 40°C/75% RH and 30°C/65% RH	Stability chamber	Assay, dissolution, appearance, moisture at 0, 1, 3, 6 months	"Executed ICH accelerated and long-term stability studies"
12	Particle Size Analysis — Measure API PSD	Malvern or sieve analysis	D10, D50, D90, span	"Characterized API particle size distribution using laser diffraction"
13	Flow Property Assessment — Carr's Index, Hausner Ratio, Angle of Repose	Tapped density apparatus, funnel	Carr's Index, Hausner Ratio, flow classification	"Evaluated powder flow properties using compendial methods"
14	Capsule Formulation — Fill hard gelatin capsules with optimized blend	Manual capsule filler	Weight variation, disintegration, dissolution	"Formulated and evaluated hard gelatin capsules"
15	ODT Preparation — DC with mannitol + high crospovidone	Press, disintegration apparatus	Disintegration time (<30 sec), wetting time, dissolution	"Developed orally disintegrating tablets with <20 sec disintegration using direct compression"

5.2 Data Documentation for CV

Professional Graph Types to Generate

1. **pH-Solubility Profile:** Line graph with error bars (pH on x-axis, solubility on y-axis)
2. **DSC Overlay Thermograms:** Pure API vs API+excipient mixtures (endothermic peak shifts = incompatibility)
3. **Dissolution Profiles:** % Drug released vs time; multiple formulations overlaid; error bars (n=6)
4. **DoE Contour Plots / 3D Surface Plots:** From JMP or Design-Expert

5. **Release Kinetics Plots:** Zero-order, first-order, Higuchi, Korsmeyer-Peppas model fits
6. **HPLC Chromatograms:** Clean peak with labeled retention time, resolution from degradants
7. **Forced Degradation Summary:** Bar chart showing % degradation under each stress condition
8. **Stability Trend Graphs:** Assay (%) vs time (months) for each storage condition

Statistical Analysis to Perform

- **ANOVA** (one-way) for comparing formulation batches
- **t-test** for paired comparisons
- **Regression analysis** for calibration curves ($R^2 > 0.999$)
- **f1/f2 calculation** for dissolution comparison
- **Release kinetics** fitting (R^2 for each model)
- **DoE analysis** — ANOVA for model significance, contour plots

5.3 CV Presentation of Formulation Work

Bullet Point Format: **Action Verb + Technique + Quantified Result**

India CV Examples:

- "Designed and optimized sustained-release matrix tablets of [Drug] using HPMC K100M by QbD approach, achieving 8-hour controlled release ($f_2 = 58$ with innovator)"
- "Developed and validated stability-indicating RP-HPLC method for [Drug] per ICH Q2(R2) ($R^2 = 0.9998$, RSD $< 1.5\%$)"
- "Conducted excipient compatibility screening using DSC and FTIR for 8 excipients, identifying and resolving a critical incompatibility with lactose monohydrate"
- "Applied Box-Behnken DoE (3 factors, 15 runs) using Design-Expert to optimize polymer concentration and compression parameters"
- "Executed ICH Q1A accelerated stability studies at $40^\circ\text{C}/75\% \text{ RH}$; product stable for 6 months (no significant change in assay or dissolution)"

USA/EU CV Examples (more result-oriented):

- "Led formulation development of an extended-release oral solid dosage form from preformulation through pilot-scale (5kg) manufacture, achieving target dissolution profile ($f_2 > 55$ vs RLD)"
- "Applied QbD principles: defined QTPP, identified 5 CQAs, conducted FMEA risk assessment, and established Design Space using Box-Behnken DoE (Design-Expert 13)"

- "Developed stability-indicating analytical method and executed forced degradation study under 5 ICH stress conditions, resolving 4 degradation products with Rs >2.0"

5.4 Project Topics That Impress Companies

IN India (Sun Pharma, DRL, Cipla, Lupin) — 10 Topics

#	Topic	Why It Impresses
1	QbD-based development of extended-release tablets of Metformin HCl using HPMC matrix	Core generic product; demonstrates QbD competency
2	Solubility enhancement of a BCS Class II drug using solid dispersions (HME or co-precipitation)	Hot topic; relevant to complex generics pipeline
3	Formulation and evaluation of gastroretentive floating tablets of Ciprofloxacin HCl	Classic Indian thesis topic; demonstrates dosage form understanding
4	Development of taste-masked ODT for Ondansetron using ion exchange resin	Practical skill; ODT is growing market
5	SNEDDS formulation of Fenofibrate with in vitro lipolysis evaluation	Lipid formulation expertise; relevant to global filing
6	Optimization of enteric-coated pellets for Omeprazole using Wurster coating	Multiparticulate expertise; highly relevant to Indian generics
7	Nanocrystal formulation of Itraconazole by wet media milling	Nanotechnology + practical relevance
8	Formulation of bilayer tablets combining IR and SR layers for Metoprolol	Bilayer technology is commercially important
9	Transdermal patch of Nitroglycerin using matrix design with permeation enhancers	TDDS is growing segment in India
10	Stability evaluation and shelf-life prediction of a generic tablet product using Arrhenius approach	Demonstrates regulatory readiness

us USA (Pfizer, Merck, Lilly, AbbVie) — 10 Topics

#	Topic	Why It Impresses
1	Amorphous solid dispersion of a BCS II drug via spray drying with HPMC-AS: supersaturation and precipitation inhibition	State-of-art platform technology

#	Topic	Why It Impresses
2	Hot-melt extrusion of a poorly soluble compound: process optimization and scale-up evaluation	HME expertise highly valued
3	QbD-driven development with full QTPP, CQA, DoE, and Design Space documentation	Demonstrates FDA-ready thinking
4	IVIVC (In Vitro-In Vivo Correlation) development for a modified-release formulation	Advanced biopharmaceutics; rare in M.Pharm theses
5	Lipid nanoparticle formulation for siRNA delivery: optimization of ionizable lipid ratio	Cutting-edge; LNP is highest-demand area
6	Continuous manufacturing of oral solid dosage forms: twin-screw granulation + compression	Industry 4.0 theme; FDA encourages continuous manufacturing
7	Abuse-deterrent formulation using PEO matrix with physicochemical evaluation	Opioid crisis → abuse-deterrent is major USA focus
8	3D-printed personalized dosage forms for pediatric patients	Future-oriented; FDA has approved first 3D-printed drug (Spritam)
9	Cyclodextrin-based solubilization: comparative evaluation of HP-β-CD and SBE-β-CD	Practical knowledge of two key platforms
10	Long-acting injectable PLGA microsphere formulation: release mechanism and sterilization evaluation	High-value injectable market

EU EU (Novartis, Roche, AstraZeneca, Sanofi) — 10 Topics

#	Topic	Why It Impresses
1	Patient-centric pediatric formulation: mini-tablets or orodispersible film	EU Paediatric Regulation drives this
2	Green chemistry approach to formulation: solvent-free HME vs spray drying	Sustainability is major EU pharma theme
3	QbD with PAT integration: in-line NIR monitoring of blend uniformity	Process analytical technology is EU regulatory expectation
4	Biosimilar formulation development: mAb stability in different buffer systems	Biosimilar is huge EU market
5	Nanoparticle drug delivery for cancer: PEGylated PLGA nanoparticles with targeting ligand	Oncology R&D is dominant in EU

#	Topic	Why It Impresses
6	Regulatory comparison: development of a product dossier for simultaneous EU/USA filing	Demonstrates regulatory fluency
7	Stability prediction using AI/ML models on pharmaceutical degradation data	Digital pharma; EU AI Act context
8	Inhalation formulation: DPI (dry powder inhaler) development with lactose carrier optimization	Pulmonary is strong EU therapy area
9	Hydrogel-based wound dressing with sustained antimicrobial release	Medical device/drug combination
10	Co-crystal screening and characterization for a poorly soluble API	Thermodynamically stable approach (vs ASD)

Tip

Master Tip for All Markets: Whatever your thesis topic, ensure you can explain it using the QbD framework (even if you didn't formally use QbD). Frame your API's properties as preformulation data, your formulation variables as CPPs/CMAs, your quality tests as CQAs, and your optimization as a DoE-driven approach. This language resonates universally with pharma R&D interviewers in India, USA, and EU.

This document, combined with Part 1, forms a comprehensive reference for mastering pharmaceutical formulation science at the R&D Scientist interview level.